

Confidence_Interval_1proportion_WVC_Jacknich Statistics Activity

Follow the Colab Notebook instructions to load the data and start exploring.

Q1. What is the population this sample is drawn from?

Q2. What kind of study is this? Experimental or observational? And why is this important?

Calculate a single confidence interval.

-Q3. Write your confidence interval in the format of (lower bound, upper bound)

-Q4. This sample data set has flight information from 2014 between SF bay area and Seattle, Washington. Write a sentence, or two, to interpret the confidence interval you found in context.

-Q5 What is the population proportion of flights between the bay area and Seattle that was delayed at least 15 minutes on take off?

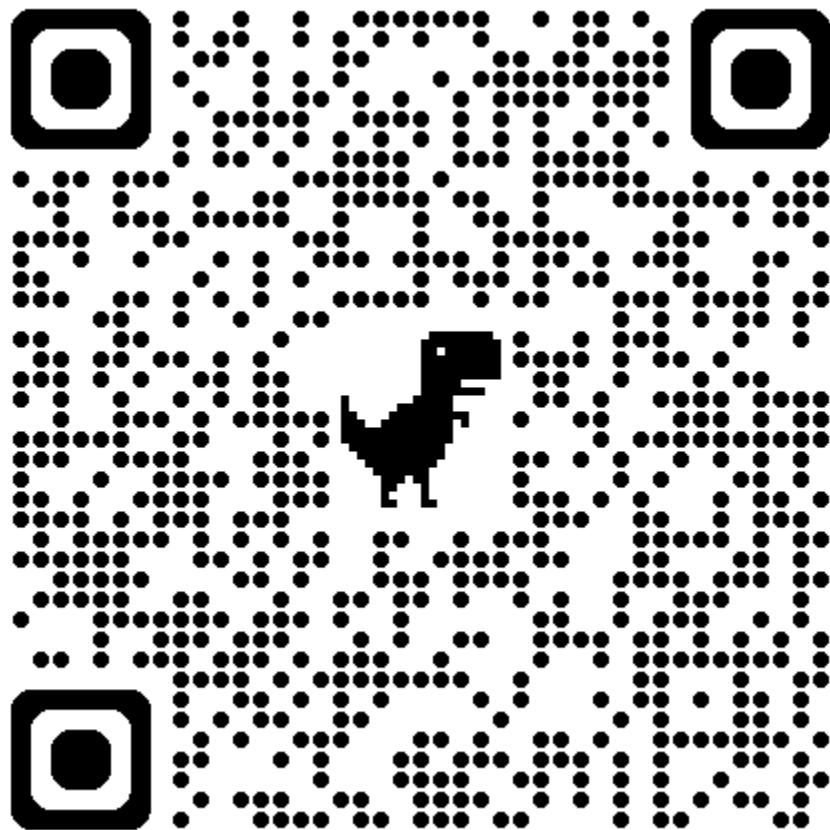
-Q6 Now that you know the true population proportion, did your confidence interval (built using a random sample and $\hat{p}$) contain the population proportion, p ? If p was included in the confidence interval, was it in the middle?

Calculate a 100+ confidence interval to understand “C-level”.

-Q7 If we drew 200 random samples and constructed 200 CIs with these samples using a confidence level of 0.95, how many CIs do you expect to capture p ?

-Q8 Why do you think some CIs do not capture p ?

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Students need to sign into a google account to run the cells in this activity.